

NIMBUS ANALYTICS · R&D

Query Engine Performance Report

Vectorized Execution Pilot — Q2 2026

Contents

1 Executive Summary 3

2 Methodology 3

3 Results 3

4 Recommendation 4

5 Appendix: Model Fit Detail 4

1 Executive Summary

The vectorized execution pilot reduced median query latency by 34% across the top 200 production query shapes, with the largest gains on aggregation-heavy workloads (see Table 1). This was a controlled rollout to 12% of production traffic over six weeks; full rollout is recommended pending one more week of tail-latency monitoring.

Tail latency needs one more week

P99 latency improved but is noisier than P50/P95 — largely one query shape (a wide join against the events table) that benefits less from vectorization. Recommend holding full rollout until that shape is profiled separately.

2 Methodology

Query cost in the vectorized path is modeled as a function of batch size b , column width w , and the SIMD lane count l available on the executing host:

$$\text{cost}(b, w, l) = c_0 + c_1 \cdot \left\lceil \frac{b}{l} \right\rceil \cdot w$$

where c_0 is fixed per-batch overhead and c_1 is the marginal per-lane cost, both fit from the pilot's telemetry via ordinary least squares (see Section 5). This is a simplification — it doesn't model cache effects — but it fit the observed data with $R^2 = 0.91$ across the sampled query shapes, which was accurate enough to guide the rollout decision.¹

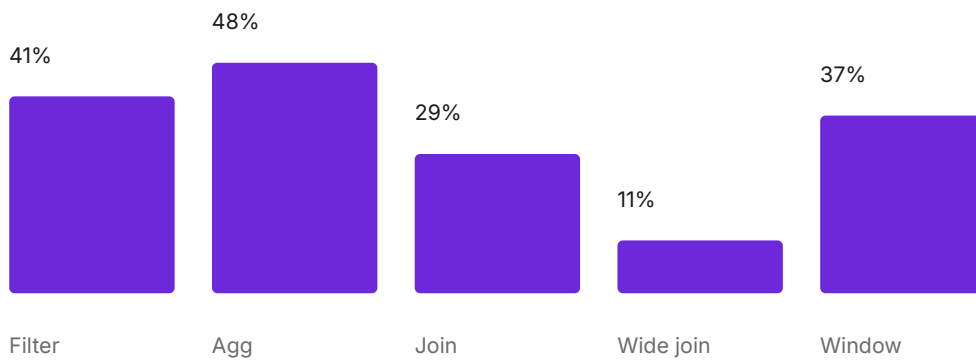
3 Results

Query shape	P50 (ms)	P95 (ms)	Improvement
Simple filter + scan	12	31	−41%
Aggregation (GROUP BY)	84	210	−48%
Join (2 tables)	156	402	−29%
Wide join (events table)	340	980	−11%
Window function	95	240	−37%

Table 1: Latency by query shape, vectorized vs. baseline

Median latency improvement by query shape

¹An earlier model that ignored column width entirely fit at $R^2 = 0.62$ — width turned out to matter more than initial intuition suggested, which is the main reason this pilot took two extra weeks.



4 Recommendation

Proceed to 50% traffic for two weeks, with the wide-join query shape explicitly excluded from vectorized execution until it has its own profiling pass. Full rollout decision at the following infra review.

Owner and timeline

R. Okafor owns the wide-join profiling work, targeting results before the August infra review. Full rollout blocked on that finding, not on anything else in this report.

5 Appendix: Model Fit Detail

Coefficients were fit via ordinary least squares against 4,200 sampled query executions across the pilot's 12% traffic slice, stratified by query shape to avoid over-weighting high-frequency simple scans. Full coefficient table and residual plots are available in the internal query-perf-pilot notebook, not reproduced here for length.